

Tech & Teach

Digital Chip Design Program

RTL Design

Task #01

Farah Alyahya

Rana Alsubaihi

Joud Almshari

Tala Alrsheed

Hoor Alkhedairi

Horiah Alqufaydi

30/6/2026

Gate - level

Version 1.3

Task #1.1

Project Summary

Schematic

task1.sv *

?

□

□

C:/Task01/Task01.srcs/sources_1/new/task1.sv

🔍

📁

↶

↷

✂

📄

📄

✖

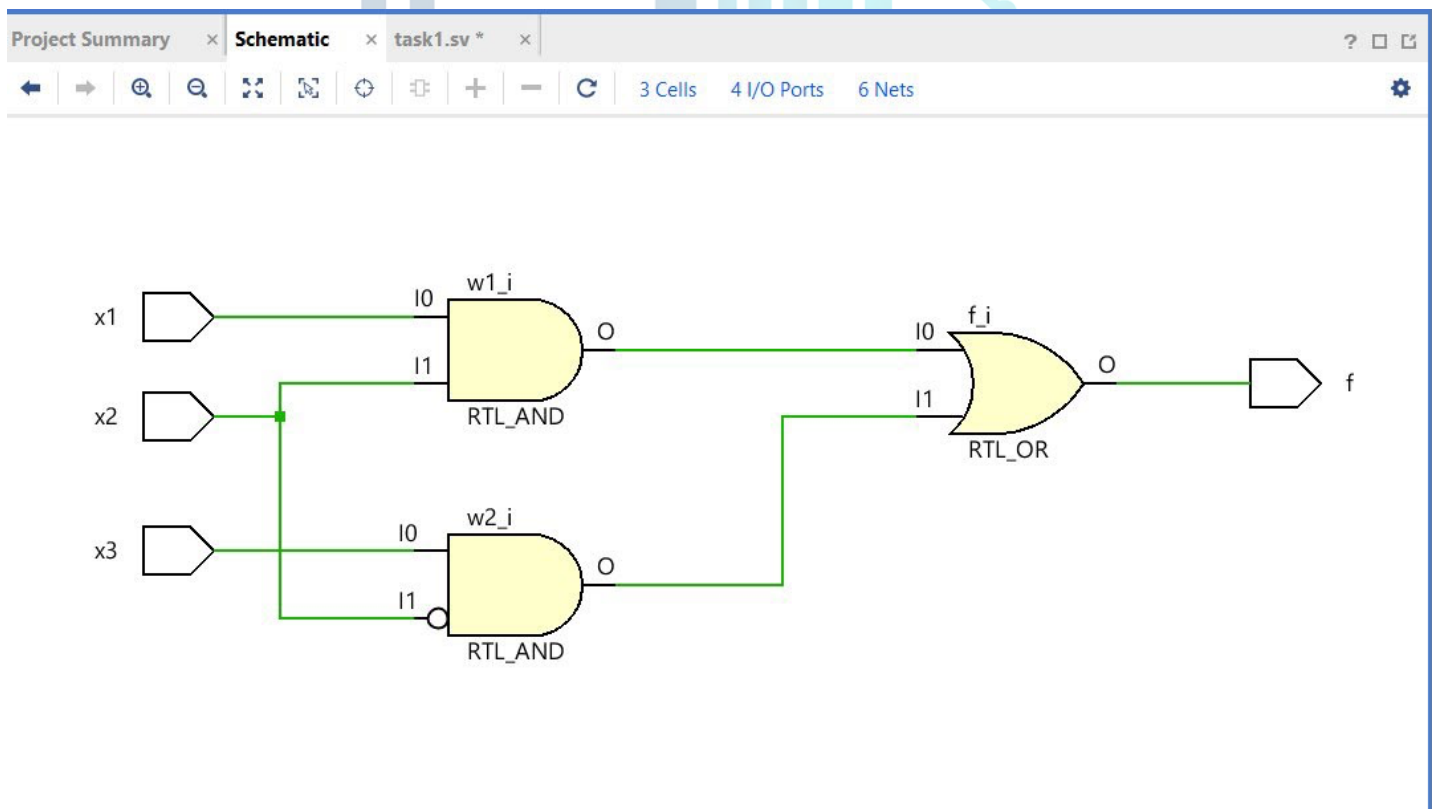
//

📊

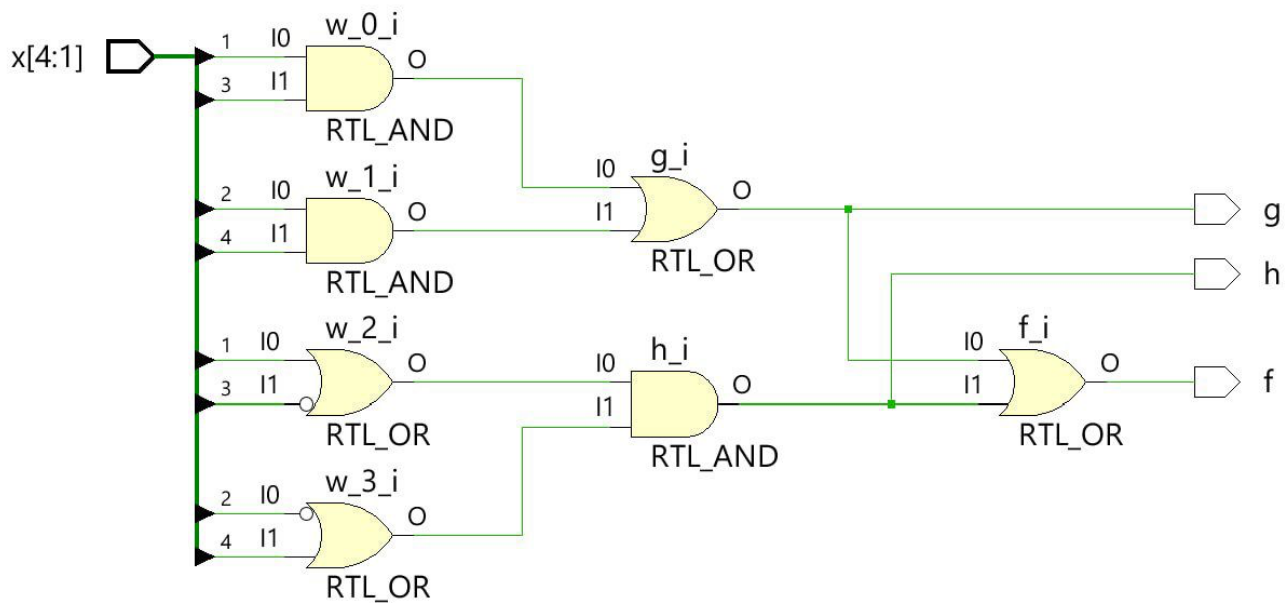
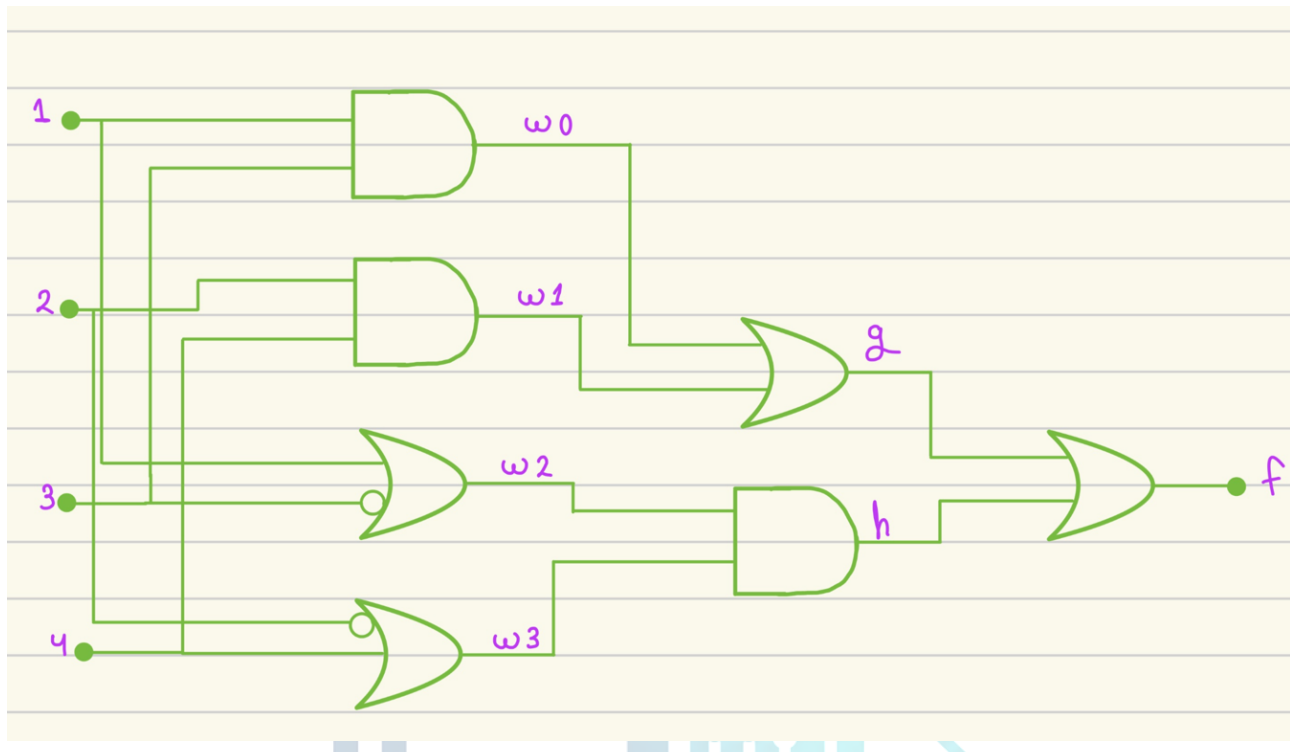
💡

⚙

```
1  `timescale 1ns / 1ps
2
3
4
5  module task1(
6      input logic x1,
7      input logic x2,
8      input logic x3,
9      output logic f
10     );
11
12     logic w0,w1,w2;
13     and(w1,x1,x2);
14     not(w0,x2);
15     and (w2,x3,w0);
16     or (f,w1,w2);
17 endmodule
18
```

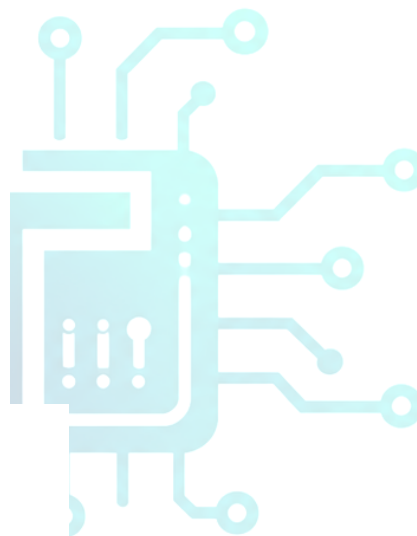


Task #1.2



Task #1.3

```
1 module Day1Task(  
2     input logic a3, a2, a1, a0,  
3     input logic b3, b2, b1, b0,  
4     output logic AeqB,  
5     output logic AltB,  
6     output logic AgtB  
7 );  
8  
9     wire i3, i2, i1, i0;  
10    wire nb3, nb2, nb1, nb0;  
11    wire gt3, gt2, gt1, gt0;  
12  
13    xnor (i3, a3, b3);  
14    xnor (i2, a2, b2);  
15    xnor (i1, a1, b1);  
16    xnor (i0, a0, b0);  
17  
18    // AeqB = i3 i2 i1 i0  
19    and (AeqB, i3, i2, i1, i0);  
20  
21    // NOT for B bits  
22    not (nb3, b3);  
23    not (nb2, b2);  
24    not (nb1, b1);  
25    not (nb0, b0);  
26  
27    // AgtB logic  
28    and (gt3, a3, nb3);  
29    and (gt2, i3, a2, nb2);  
30    and (gt1, i3, i2, a1, nb1);  
31    and (gt0, i3, i2, i1, a0, nb0);  
32  
33    or (AgtB, gt3, gt2, gt1, gt0);  
34  
35    // AltB = NOT(AgtB OR AeqB)  
36    nor (AltB, AgtB, AeqB);  
37  
38 endmodule
```



Teach

Page 5 of 5